

SAT-SA

Supervisory Analytics Tool for SOC Assessment

Architecture Document

Offline, Evidence-Driven Supervisory Analytics for Critical Sector Entities

Executive Summary

SAT-SA (Supervisory Analytics Tool for SOC Assessment) is an evidence-driven analysis platform that processes periodic SOC submission data from Critical Sector Entities (CSEs) to surface execution gaps, negative-space anomalies, and other supervisory signals.

It ingests local CSV/JSON data, applies deterministic detectors for execution gaps, silent assets, anomalous trends, computes a Strategic Risk Index (SRI), and presents findings with transparent thresholds and supporting evidence. The interface highlights high-attention entities for examiner review and supports report export.

1. System Architecture

The SAT-SA processing pipeline converts submitted CSE data into explainable supervisory findings while retaining the examiner in the decision loop.

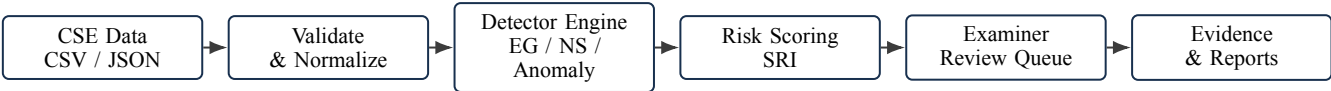


Figure 1. SAT-SA system architecture and supervisory data flow.

2. Detector Catalogue

Key detectors implemented in SAT-SA include:

ID	Short Description	Category
EG-01	Critical alerts without escalation	Execution Gap
EG-02	Silent critical assets	Execution Gap
NS-01	Missing security telemetry	Negative Space
NS-02	Low alert density	Negative Space
AN-01	Repeated investigations pattern	Anomaly
AN-02	Rapid-closure alert spike	Anomaly

Example Finding — EG-01:

EG-01 - Critical Alert Without Escalation
Trigger: severity = CRITICAL AND escalated = NO
Observed: 7 alerts
Evidence: AL-1021, AL-1083, AL-1107, ...

This finding is shown with its detector logic, threshold, observed value, and supporting evidence so the examiner can see why it was flagged.

3. Supervisory Analytics

SAT-SA analyses operational evidence across entities and time to support supervisory sampling:

- Execution Gaps:** Behavior inconsistent with expected operational controls, including missing escalations and closures.
- Negative Space:** Expected evidence that is absent, such as silent assets, missing telemetry or unusually low activity.
- Peer Benchmarking:** Entity behavior compared against peer cohorts for deviations.
- Anomaly Detection:** Unusual operational patterns and outliers.
- Supervisory Prioritisation:** Entity and finding priorities ranked for examiner review.

4. Explainability and Human Review

SAT-SA does not make final decisions; it equips the examiner with clear evidence. Each finding displays:

Element	What is shown to the examiner
Detector	Stable detector ID and description
Trigger	Logic formula that triggered the finding
Threshold	Configured detector threshold
Observed	Calculated value from submitted data
Evidence	Alert/case IDs and supporting records
Peer Context	Comparable peer metrics if available
Examiner Action	Corroborates / Does Not Corroborate / Needs More Evidence

This preserves the human-in-the-loop model: the tool identifies signals; the examiner makes the supervisory judgment.

5. Validation and Performance

- **Synthetic gold-set recovery:** 21/21 expected signals recovered in prototype validation.
- **Performance:** 134k alerts across 720 entities processed in approximately 27 seconds (benchmarked).
- **Peer analytics:** Peer thresholds and comparisons are exposed for transparency.
- **Limitation:** Synthetic gold-set shows detector recovery; real-world efficacy needs testing against labeled outcomes.

6. Offline Deployment

SAT-SA is built for an air-gapped environment:

Requirement	Approach
Internet	Not required during runtime
Cloud/SaaS	No external dependency
External AI	None used
Processing	Local (client-side) only
Deployment	Workstation or internal network
Runtime	Browser interface on local machine

7. Security and Auditability

The prototype provides traceable controls:

- SHA-256 hashing for tamper-evident inputs/outputs.
- Append-only logs of examiner actions and timestamps.
- Signed reports for integrity of exported assessments.
- Deterministic detector logic for reproducible runs.

8. Examiner Demonstration Workflow

1. Launch the SAT-SA local interface on the workstation.
2. Verify offline status and current data corpus (e.g., via status indicators).
3. Open **Command Brief** to identify entities needing review.
4. Open **Findings** and inspect an execution gap or negative-space finding.
5. View the evidence drawer to see underlying alert/case details.
6. Open **Review Queue** and record the examiner's judgment.
7. Use **Methodology** to inspect detector logic and validation evidence.
8. Export the final assessment report or JSON for audit.

SAT-SA assists the examiner; it does not replace the expert's judgment.